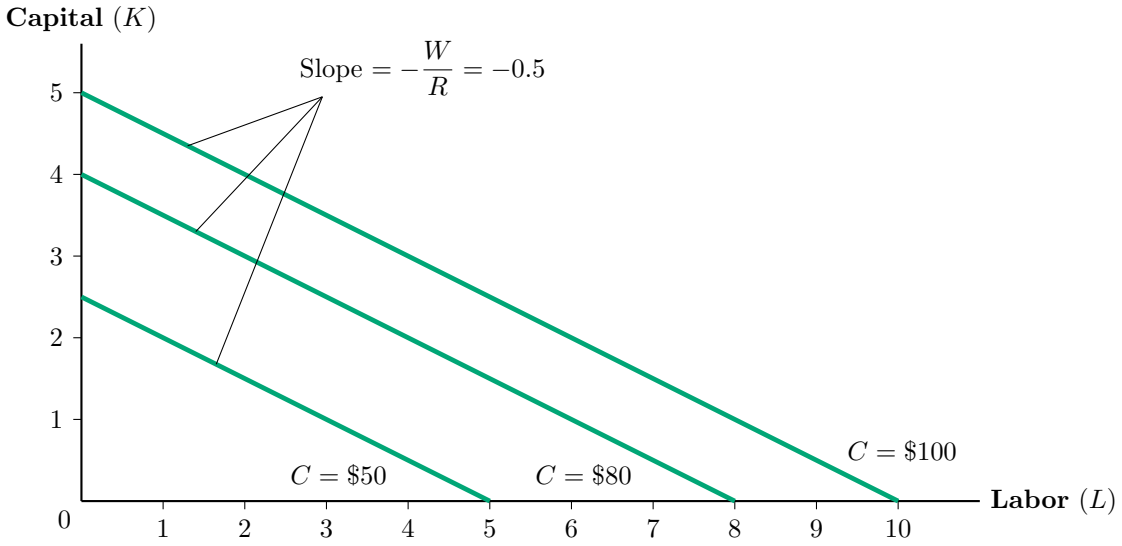
Class 14 Quiz Solutions

PUBPOL 205

October 15, 2026

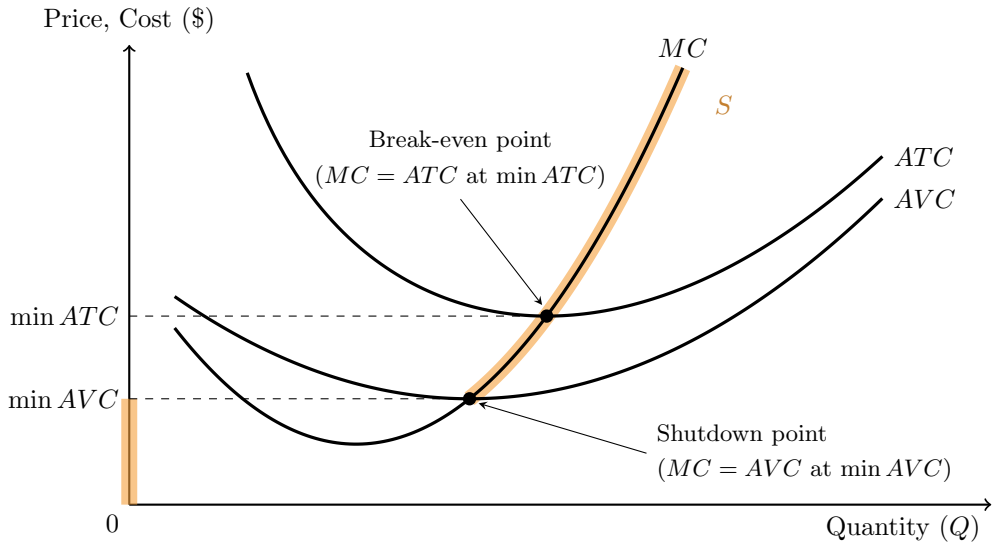
1. The graph below shows three isocost lines for a firm that hires labor (L) at wage W and rents capital (K) at rental rate R. Is labor more expensive, or is capital more expensive? Why?



Adapted from Goolsbee et al., *Microeconomics*, 3e.

> **Solution. Capital is more expensive.** An isocost line is the set of bundles that cost the same total amount, $C = WL + RK$. Solving for K:
>
> $$K = \frac{C}{R} - \frac{W}{R} L.$$
>
> The slope is minus the price of labor relative to the price of capital. Here $-W/R = -0.5$, so $W = 0.5R$: one unit of labor costs half as much as one unit of capital. Equivalently, one unit of capital costs as much as two units of labor. Along an isocost line, hiring one more unit of labor at the same total cost means giving up half a unit of capital.
> Check with the intercepts of the $C = \$100$ line: \$100 buys 10 units of labor, so $W = \$10$, or 5 units of capital, so $R = \$20$. Then $W/R = 10/20 = 0.5$.

2. Consider a perfectly competitive firm in the short run. On the axes below, draw the firm's marginal cost (MC), average variable cost (AVC), and average total cost (ATC) curves, and sketch out the firm's short-run supply curve. Where is the break-even point? Where is the shutdown point? Label both on your graph.



Solution.

- **The curves.** MC crosses both AVC and ATC at their minimum points: when MC is below an average, one more unit pulls the average down, and when MC is above it, one more unit pulls it up. Formally, $d(VC/Q)/dQ = (MC - AVC)/Q$, which is zero exactly where $MC = AVC$ (same for ATC). ATC lies above AVC by $AFC = FC/Q$, which shrinks as Q grows, so the curves get closer and min ATC lies above and to the right of min AVC.
- **Break-even point:** where $MC = ATC$, at the minimum of ATC. At $P = \min ATC$ the firm produces where $P = MC = ATC$ and earns zero economic profit; above this price it earns a profit, below it a loss.
- **Shutdown point:** where $MC = AVC$, at the minimum of AVC. If $P < \min AVC$, revenue does not cover variable cost, so producing loses more than shutting down (which loses only FC). The firm produces $Q = 0$.
- **Short-run supply curve** (shaded): the firm sets $P = MC$, so supply is MC above min AVC, and $Q = 0$ below min AVC.
- **Between the two points** ($\min AVC < P < \min ATC$), the firm keeps producing at a loss: revenue covers all variable cost and part of fixed cost, so the loss is smaller than FC. Example: $FC = \$40$, $P = \$8$, and at the output where $P = MC$, $Q = 10$, $AVC = \$6$, $ATC = \$10$. Producing gives profit $(8 - 10) \times 10 = -\$20$; shutting down gives $-\$40$. Produce. (At exactly $P = \min AVC$ the loss is FC either way, so the firm is indifferent.)